

Inspection and Classification System of Photovoltaic Module Defects Based on UAV and Thermal Imaging

Xin Zhang
Suzhou Nuclear Power Research
Institute Co., Ltd.
Suzhou, China
zhangxin20162@cgnpc.com.cn

Pingguo Zou
Suzhou Nuclear Power Research
Institute Co., Ltd.
Suzhou, China
zoupingguo@cgnpc.com.cn

Chi Ma
CGN New Energy Holdings Co.,
Ltd.
Beijing, China
machi@cgnpc.com.cn

Zhentao Zhang
Suzhou Nuclear Power Research
Institute Co., Ltd.
Suzhou, China
zhangzhentao@cgnpc.com.cn

Hongbin Guo
CGN New Energy Holdings Co., Ltd.
Beijing, China
guohongbin@cgnpc.com.cn

Yabin Chen
Suzhou Nuclear Power Research Institute
Co., Ltd.
Suzhou, China
chenyabin2@cgnpc.com.cn

Zikun Cheng
Suzhou Nuclear Power Research Institute
Co., Ltd.
Suzhou, China
chengzikun@cgnpc.com.cn

Abstract—With the vigorous development of large-scale photovoltaic power plants, the demand and requirements for defect inspection of photovoltaic power plant modules are also increasing. At present, manual inspection is inefficient, costly and insecure. The inspection method combined with UAV and thermal imaging technology has high efficiency, low cost and less safety risk, which is an effective means to realize unmanned operation and maintenance of photovoltaic power plants. Combined with the image recognition technology of artificial intelligence, this paper designed an inspection and classification system based on UAV. Firstly, this paper summarized the corresponding relationship between the thermal image characteristics and the common defects of photovoltaic modules, and established a knowledge base of photovoltaic module defects based on image features. Then, an automatic image recognition algorithm model of photovoltaic modules based on YOLOV5 is built. Finally, based on the UAV and thermal camera, an inspection and classification system is built for testing and verification. The system collects thermal images of photovoltaic modules by UAV, and then distinguishes thermal anomalies of different shapes by AI automatic identification technology. The defects such as shelter, smudge, bypass diode damage, open circuit and short circuit are identified, and the treatment measures are given. On the basis of ensuring the recognition accuracy, the system greatly improves the efficiency of photovoltaic module detection, and has been successfully applied in a photovoltaic power plant.

Keywords—photovoltaic module defects, thermal inspection, UAV

I. INTRODUCTION

With the large-scale construction of photovoltaic power plants, the ability to detect and identify defects in photovoltaic modules is crucial to forming a reliable, efficient and safe power generation system. In this way, it is possible to avoid energy losses and safety problems[1].

Affected by the photovoltaic module manufacturing process, transportation process, and installation process, the photovoltaic modules to be put into operation have various defects. Although, in the process of equipment manufacturing and equipment installation, equipment inspectors are able to discover modules defects to a certain extent through visual inspection, I-V curve test and manual EL test. However, visual inspection, I-V curve test and manual EL consume a lot of time and manpower[2-3]. If sampling inspection is used, it cannot fully reflect the quality status of all photovoltaic modules. It is necessary to develop a more efficient method of finding and identifying various defects in photovoltaic modules. Defects can cause various abnormalities in the thermal imaging of photovoltaic modules compared to normal operating photovoltaic modules. It is a relatively advanced detection method to use an unmanned aerial vehicle (UAV) equipped with a thermal camera to take a thermal image of the photovoltaic modules, and to find defective modules through the images. In particular, this detection method is time-saving, non-destructive, and non-contact, which can effectively improve the detection efficiency of large-scale photovoltaic power plants[4-5].

This article expounds the corresponding relationship between thermal imaging of photovoltaic modules and module defects, and establishes a knowledge base of photovoltaic module defects based on thermal imaging features. An inspection and classification system based on UAV and thermal imaging is designed to inspect photovoltaic modules. Use UAV and thermal camera to collect thermal images of photovoltaic modules in photovoltaic power plants. Using the target detection algorithm based on YOLOV5, the thermal imaging of photovoltaic modules is analyzed, and the thermal abnormal area is found. Combined with the defect knowledge base, the types of defects are identified. This method can cover 100% of installed photovoltaic modules, and improve the efficiency of defect detection of photovoltaic power plant. At the same time, it also overcomes the hidden trouble of

operation safety in the process of finding defects by manual inspection.

II. CLASSIFICATION OF MODULE DEFECTS BY THERMAL IMAGES

There are various forms of abnormality in the thermal image of photovoltaic modules. Only by finding the cause of the abnormality and distinguishing the defect types corresponding to different thermal images can we provide effective suggestions for the treatment of defects. By analyzing the mechanism of thermal anomaly of photovoltaic modules, the corresponding relationship between common defects of photovoltaic modules and thermal anomaly images is expounded. A knowledge base of photovoltaic module defects based on image features is established.

A. Hot Spot Abnormality

The Hot spot abnormality is shown in Figure 1. The temperature of some areas on the photovoltaic module increases abnormally, showing a spot shape. This is the most common photovoltaic module thermal abnormality. The reason for the abnormality is that one or several solar cells in the module fail or the photovoltaic module is covered with stains such as bird droppings and dust. Breakage of front glass can also cause hot spot abnormality[6]. When the solar cell fails, the current generated by the cell is less than the circuit current, the cell bears the negative voltage, becomes the load in the circuit, and eventually consumes the energy generated by other solar cells in the form of heat.

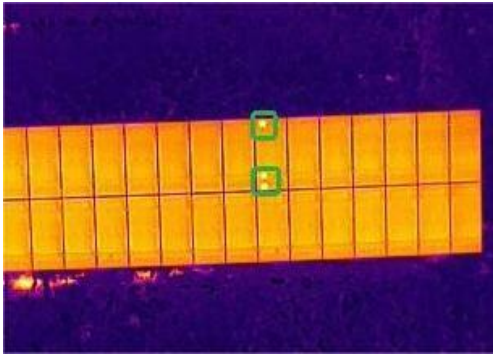


Fig. 1. Hot spot abnormality

B. Sub-string Abnormality

In a photovoltaic module, multiple cells are connected in series to form photovoltaic sub-strings, and a photovoltaic module is usually composed of several photovoltaic sub-strings. The Sub-string abnormality is shown in Figure 2, the temperature of one sub-string is higher than other sub-strings in the same photovoltaic module, forming a striped bright spot. The reason is that the bypass diode is turned on, the sub-string is disconnected by the bypass or the sub-string is falsely welded in the junction box. Bypass diodes are an integral part of photovoltaic modules. The bypass diode divides the photovoltaic modules into different sub-strings. When the solar cell in the sub-string fails, the bypass diode will connect and disconnect the corresponding sub-string bypass. In addition, the bypass diode is turned on due to its own fault or the sub-string loses the effective connection in the junction box, the

sub-string will lose its contribution to the power generation of the module. In these cases, the light energy on the sub-string cannot be converted into electric energy, heating the sub-string so that the sub-string temperature is higher than that of the other sub-strings in the module.

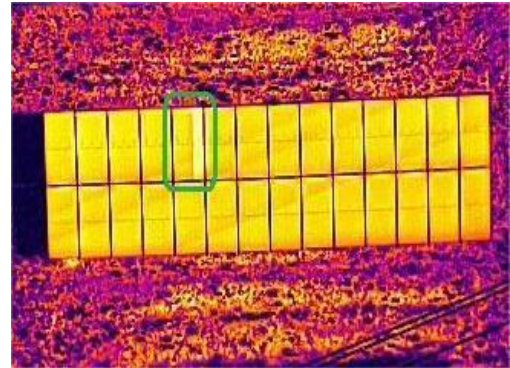


Fig. 2. Sub-string abnormality

C. Module string abnormality

In a photovoltaic power plant, multiple photovoltaic modules are connected in series to form a module string. The whole module string will also lead to thermal abnormalities due to different defects. Module string heat anomaly can be divided into open circuit and short circuit. As the open circuit abnormality shows in Figure 3, the temperature of all photovoltaic modules in a module string are higher than other surrounding modules, and the photovoltaic modules warm evenly. The reason is that the module string is open-circuited, not connected to the power generation system. The light energy on the modules cannot be converted into electricity, heating the modules so that the temperature are higher than other normally generating modules.

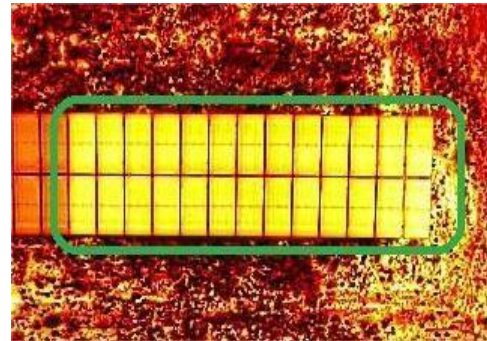


Fig. 3. Open circuit abnormality

The short circuit abnormality is shown in Figure 4, the temperature of the photovoltaic modules in a module string are higher than other surrounding modules, and the photovoltaic modules warm not evenly. The reason is that the module string is short-circuited, and the electric energy converted by light energy forms a current path inside the module string, which produces a thermal effect on the internal resistance of the modules in the string, causing an uneven temperature rise on the modules.

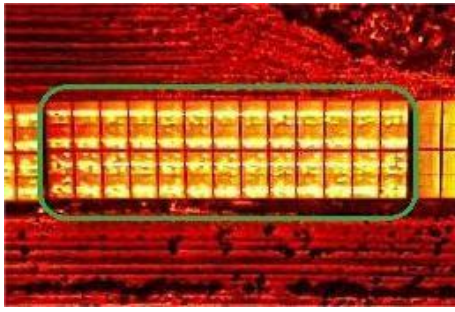


Fig. 4. Short circuit abnormality

D. Other Image Abnormality

According to the images of photovoltaic modules, in addition to defects that can be found from thermal abnormality, defects can also be found through the shape features of the images. As shown in Figure 5, the overturning or shedding of the modules leads to the abnormal position of some modules, which is generally caused by extreme weather such as strong wind and heavy snow.

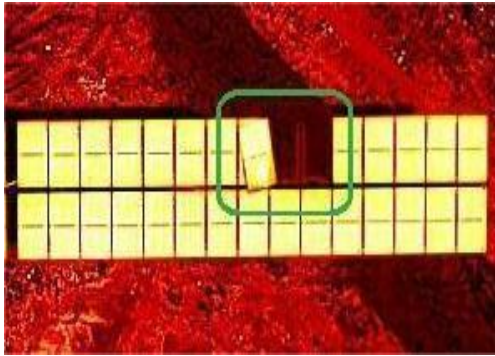


Fig. 5. Module position abnormality

As shown in Figure 6, there are large shadows on the surface of the photovoltaic modules, usually caused by vegetation or buildings or transmission towers near the photovoltaic modules. This kind of shading is also a typical defect of photovoltaic modules in actual operation.

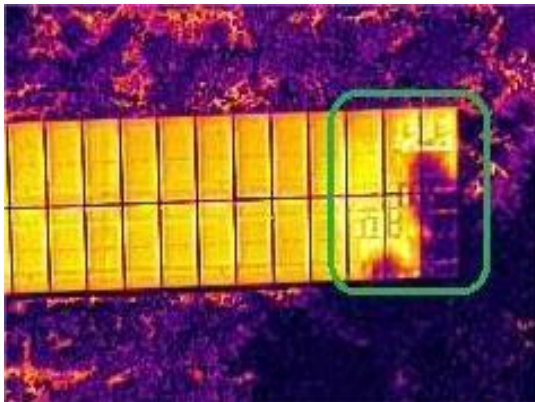


Fig. 6. Shadow abnormality

III. INSPECTION SYSTEM OF THERMAL IMAGING BASED ON UAV

The inspection and classification system uses the UAV equipped with a thermal camera to collect thermal images of photovoltaic modules to realize efficiency and safety inspection of photovoltaic power plant modules. The system uses the YOLOv5 algorithm to automatically identify the thermal image abnormality of photovoltaic modules, and then determines the defect type corresponding to the abnormality through comparative analysis with the knowledge base.

A. Image Target Detection Algorithm Based on YOLOv5

The algorithm based on convolutional neural network is the mainstream algorithm in the field of artificial intelligence target detection, among which YOLO series algorithm has simple model, fast detection speed and high precision, and is widely used^[7]. This system selects YOLOv5 as the algorithm for classifying and inspecting thermal anomalies of photovoltaic modules, trains the algorithm model on the self-made knowledge base of photovoltaic module defects, and uses the trained YOLOv5 algorithm to realize the automatic inspection and classification of thermal anomaly images.

The algorithm model requires the data set to complete the training and testing, and the defect knowledge base provides the indispensable data support for the implementation of the algorithm model. Infrared images of 780MW of photovoltaic modules are collected from eight photovoltaic power plants, a total of 10,772 infrared images of photovoltaic modules. Among them, 902 infrared images with typical thermal abnormality are classified and annotated, and a knowledge base of photovoltaic module defects based on image features is established. The thermal abnormality in the knowledge base are divided into six categories, which are (1)Hot spot abnormality, (2)Sub-string abnormality, (3)Open circuit, (4)Short circuit, (5)Module position abnormality, (6)Shadow abnormality.

The main hyperparameters used in this algorithm training are shown in Table I.

TABLE I.		MAIN HYPERPARAMETERS		
Learning rate	momentum	Weight decay	optimizer	epochs
0.01	0.937	0.0005	SGD	300

For the 6 types of thermal abnormality mentioned above, the algorithm model training results are shown in Figure 7. As can be seen from the figure, the algorithm model is stable after 300 iterations, The accuracy, recall and mAP value of the model are all close to 1, which indicates that the training result of the model is good.

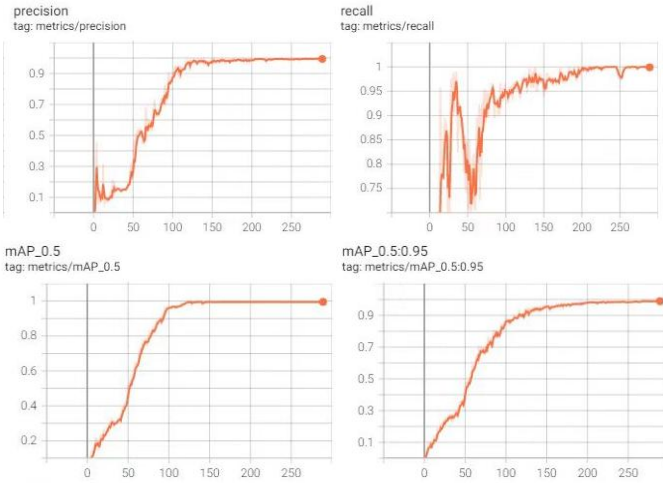


Fig. 7. Detect model training results

B. Inspection System of thermal imaging based on UAV

As shown in figure 8, the inspection and classification system is developed based on DJI M300RTK UAV and DJI H20T thermal Camera, the Main features of the equipment are shown in Table II.

TABLE II. EQUIPMENT SPECIFICATIONS

<i>Unmanned aerial vehicle</i>	
Model	DJI M300RTK
Positioning Accuracy	1 cm +1 ppm (Horizontal) 1.5 cm +1 ppm (Vertical)
<i>Thermal Camera</i>	
Model	DJI H20T
Lens	DFOV: 40.6 ° focal length: 13.5 mm aperture: f / 1.0
Scene Range	-40 °C to 150 °C (High Gain)

TABLE III. STATISTICS OF DEFECTS

<i>Defect type</i>	<i>Hot spot abnormality</i>	<i>Sub-string abnormality</i>	<i>open circuit</i>	<i>Short circuit</i>	<i>Module position abnormality</i>	<i>Shadow abnormality</i>
Number of defects	247	234	40	0	1	71
Percentage	41.7%	39.5%	6.7%	0.0%	0.2%	11.9%

Unmanned aerial vehicle

Sensitivity (NETD)	≤50 mK @ f/1.0
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Fig. 8. Inspection equipment Based on UAV and Thermal Camera

IV. SYSTEM TESTING AND VALIDATION

A. System Application

The inspection and classification system was tested at a photovoltaic power plant in Shanxi province, China. The total installed capacity of the power plant is 80MW. The test was conducted at the noon and afternoon of May 2022. The radiation value range in the detection time is 690W/m² to 1000 W/m². It was clear and cloudless. When the UAV is operating, the ambient temperature is 20 to 30 degrees Celsius, the relative height of the UAV is 65 meters, and the pitch angle of the camera is -70 degree. The inspection results are shown in Table III. In this case thermal images of 181954 photovoltaic modules were collected, identified and classified, and 593 defective modules were found in two days. Compared with manual detection, it greatly improves the efficiency and shortens the detection time.

The result of the inspection is shown in Figure 9. As can be seen from the figure, the image target detection algorithm of this system can well identify the thermal imaging anomalies of photovoltaic modules. When the confidence of the model is 0.5, the overall detection mAP of six defects of photovoltaic modules reaches 80.88%.

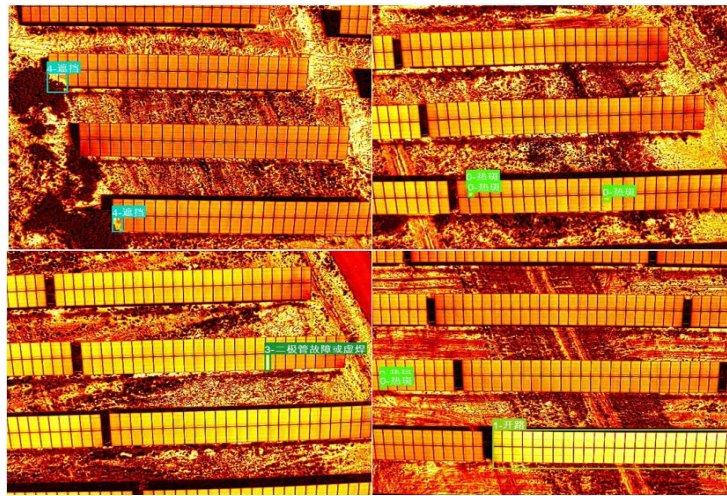


Fig. 9. Inspection result of the system

B. Defect Treatment Measures

After the UAV inspection, the operation and maintenance personnel can use the handheld thermal detector to check the defects according to the inspection results. Treatment measures for different defects are as follows.

a) Hot spot abnormality: Check the surface of the module for bird droppings, dust, fallen leaves and other shielding. If hot spots still exist after the removal of the occlusion, and the temperature difference between the hot spot and the normal module is more than 20°C, the internal fault of the cells should be considered, and the module should be replaced in time.

b) Sub-string abnormality: firstly, eliminate the surface shielding of the module, secondly, check whether the bypass diode is faulty. If the diode fails, the diode or the whole module should be replaced in time.

c) Open circuit: Check the plug of the photovoltaic modules for proper connection.

d) Short circuit: defect may cause internal damage to the photovoltaic modules, so the fault module string should be disconnected in time to find the cause of the short circuit.

e) Module position abnormality: Check whether the frame and bracket of the photovoltaic modules are damaged, and restore the defective modules in time.

f) Shadow abnormality: Check the condition of the shelter on the spot, and if it is vegetation, remove it in time.

V. CONCLUSION

The inspection and classification system proposed in this article uses the thermal images of photovoltaic modules collected by UAV, and then automatically finds and classifies the thermal anomalies of the images through YOLOv5 target

detection algorithm, so as to identify the causes of the defects of photovoltaic modules and propose treatment measures. The application in a photovoltaic power plant in Shanxi proves that the system not only greatly improves the inspection efficiency and reduces the operation and maintenance costs, but also effectively avoids the potential safety hazards of manual inspection. With the large-scale development of photovoltaic power plants, this system will have a broad market space and use value.

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